

OpenCPU SDK

PM API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Lai Hongxiao

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the PM (Power Management) APIs in the OpenCPU SDK. The main aspects covered include:

- Shutdown and Restart

Primarily used to control module shutdown or restart.

- Sleep and Wakeup

Primarily used to control module sleep and wakeup states.

- Power-On Reason

Retrieve the current reason for the module's power-on.

- Set Voltage Level

Configure the voltage level for corresponding peripherals.

- Peripheral Power Control

Turn on or off the power supply for specific peripheral pins.

2 Data Structure

2.1 nwy_pm_id_e

```
typedef enum
{
    NWY_POWER_LCD,
    NWY_POWER_KEYLED,
    NWY_POWER_BACK_LIGHT,
    NWY_POWER_SD,
    NWY_POWER_CAMA,
    NWY_POWER_CAMD,
    NWY_POWER_SIM1,
    NWY_POWER_MAX
}nwy_pm_id_e;
```

3 APIs

3.1 Module Shutdown or Restart

This interface function definition is located at `nwy_pm_api.h` and is used to control device shutdown or restart.

`nwy_pm_ctrl`

Format	<code>nwy_error_e nwy_pm_ctrl(int option)</code>
Description	Controls the module to perform shutdown or restart based on the provided option.
Parameter	option: Options for controlling the module's behavior: 1: Fast shutdown 2: Normal shutdown 3: Restart
Return Value	Success: <code>NWY_SUCCESS</code> Failure: <code>NWY_GEN_E_SET_FAILED</code> Invalid input parameter: <code>NWY_GEN_E_INVALID_PARA</code>

3.2 Sleep and Wakeup

This interface function definition is located at `nwy_pm_api.h` and is used to control the device to sleep or wake up.

`nwy_pm_state_set`

Format	<code>nwy_error_e nwy_pm_state_set(int mode)</code>
Description	Controls the module's sleep or wakeup states based on the mode.
Parameter	mode: Options for controlling module state: 0: <code>NWY_WAKEUP</code> - Keep awake 1: <code>NWY_ENTRY_SLEEP</code> - Sleep
Return Value	Success: <code>NWY_SUCCESS</code> Failure: <code>NWY_GEN_E_SET_FAILED</code> Invalid input parameter: <code>NWY_GEN_E_INVALID_PARA</code>

3.3 Boot Reason

This interface function definition is located at `nwy_pm_api.h` and is used to obtain the reason why the device is currently powered on.

`nwy_pm_boot_res`

Format	int nwy_pm_boot_res(void)		
Description	Retrieves the reason for the module's current boot.		
Parameter	None		
Return Value	The returned value indicates the boot reason:		
	placeholder for unknown reason	0x00000000	
	boot by power key	0x00000001	
	boot by pin reset	0x00000010	
	boot by alarm	0x00000100	
	boot by charge in	0x00001000	
	boot by watchdog	0x00010000	
	boot by wakeup	0x00100000	
	The return value may represent a combination of reasons, e.g., 0x00001010 (pin reset + charge-in).		

3.4 Set Voltage Level

This interface function definition is located at `nwy_pm_api.h` and is used to set the level of device peripherals.

`nwy_pm_level_set`

Format	<code>nwy_error_e nwy_pm_level_set(nwy_pm_id_e id, uint32_t mv);</code>
Description	Configures the voltage level of the module's peripherals.
Parameter	<code>id</code>: Corresponds to the peripheral option, based on enumeration values defined in <code>nwy_pm_id_e</code>. <code>mv</code>: The voltage level to set, defined as an unsigned integer.
Return Value	The returned value indicates the boot reason: Success: <code>NWY_SUCCESS</code> Failure: <code>NWY_GEN_E_SET_FAILED</code> Invalid input parameter: <code>NWY_GEN_E_INVALID_PARA</code>

3.5 Peripheral Power Control

This interface function definition is located at `nwy_pm_api.h` and is used to switch peripheral power.

`nwy_pm_power_switch`

Format	<code>nwy_error_e nwy_pm_power_switch(nwy_pm_id_e id, int enabled, int lp_enabled);</code>
Description	Peripheral Power Control
Parameter	<code>id</code>: Corresponds to the peripheral option, based on enumeration values defined in <code>nwy_pm_id_e</code>. <code>enabled</code>: Determines if the peripheral is enabled during non-sleep mode. 0: Disabled 1: Enabled <code>lp_enabled</code>: Determines if the peripheral is enabled during sleep mode. 0: Disabled 1: Enabled
Return Value	The returned value indicates the boot reason: Success: <code>NWY_SUCCESS</code> Failure: <code>NWY_GEN_E_SET_FAILED</code> Invalid input parameter: <code>NWY_GEN_E_INVALID_PARA</code>